

Course 5243B

5 Credits: 4

Program Elective

Source-grounded

Programming in C

- To provide a basic knowledge in C Programming.

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5243B is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 5243B.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Biomedical Engineering
Course code	5243B
Course title	Programming in C
Semester	5 Credits: 4
Course category	Program Elective
Periods per week	4 (L:3 T:1 P:0) Periods per semester : 60
Periods per semester	60

Item	Official information
Credits	4
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

22 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide a basic knowledge in C Programming.
- To enhance programming skills using Control blocks, Looping, String Manipulation, and functions to solve real world problems.
- To make capable to solve real-life programming tasks using C.

Course outcomes

CO	Official outcome	Study level
CO1	13 Applying programming. Make use of iterative control structures and arrays	See official syllabus
CO2	15 Applying in programs. Develop programs using pointers and strings to	See official syllabus
CO3	15 Applying solve problems more efficiently. Make use of functions to solve problems	See official syllabus
CO4	15 Applying effectively	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	Make use of programming concepts with C programming.	5	As specified
Module 2	Make use of iterative control structures and arrays in programs.	6	As specified
Module 3	efficiently.	5	As specified
Module 4	Make use of functions to solve problems effectively	6	As specified

MODULE 1

Make use of programming concepts with C programming.

This module is presented from the official syllabus scope for Programming in C. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Make use of programming concepts with C programming.
- Official duration: As specified
- Official topic entries captured: 5

- The full source excerpt is preserved below for coverage checking.

Understand structure of a C program 1 Understanding. Use the terms Keywords, Variables, Constants,

Understand structure of a C program 1 Understanding. Use the terms Keywords, Variables, Constants, is an official topic in Module 1 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understand structure of a C program 1 Understanding. Use the terms Keywords, Variables, Constants, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understand structure of a c program 1 understanding. use the terms keywords, variables, constants, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understand structure of a C program 1 Understanding. Use the terms Keywords, Variables, Constants, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-എന്നിരിക്കട്ടെ C പ്രോഗ്രാമിന്റെ 1
Understanding. Use the terms Keywords, Variables, Constants,
definition/meaning
.
steps, application
. Technical terms English-
.

3 Applying Data types and type qualifiers

3 Applying Data types and type qualifiers is an official topic in Module 1 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying Data types and type qualifiers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying data types and type qualifiers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying Data types and type qualifiers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-3 Applying Data types and type qualifiers

പ്രാഥമികമായി definition/meaning ഉൾപ്പെടെ. പ്രാഥമികമായി definition, steps, application ഉൾപ്പെടെ. Technical terms English-ൽ ഉൾപ്പെടെ ഉൾപ്പെടെ.

Identify output and input functions

Identify output and input functions is an official topic in Module 1 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify output and input functions using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify output and input functions should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify output and input functions means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-Identify output and input functions

പ്രതികരണങ്ങൾക്ക് definition/meaning നൽകുക. പ്രതികരണങ്ങൾക്ക് definition, steps, application നൽകുക. Technical terms English-ൽ പ്രതികരണങ്ങൾക്ക് നൽകുക.

Identify operators

Identify operators is an official topic in Module 1 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify operators using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify operators should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify operators means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഓപ്പറേറ്റർ Identify operators ഓപ്പറേറ്റർ ഓപ്പറേറ്റർ definition/meaning ഓപ്പറേറ്റർ. ഓപ്പറേറ്റർ ഓപ്പറേറ്റർ, steps, application ഓപ്പറേറ്റർ ഓപ്പറേറ്റർ. Technical terms English-ഓപ്പറേറ്റർ ഓപ്പറേറ്റർ.

Identify type casting 2 Applying Introduction to Programming concepts: Structure of a C program - Keyword, Variables, Constants, Data types and type qualifiers. Output and input functions, Operators - Arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, Assignment, Conditional and type casting.

Identify type casting 2 Applying Introduction to Programming concepts: Structure of a C program - Keyword, Variables, Constants, Data types and type qualifiers. Output and input functions, Operators - Arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, Assignment, Conditional and type casting. is an official topic in Module 1 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify type casting 2 Applying Introduction to Programming concepts: Structure of a C program - Keyword, Variables, Constants, Data types and type qualifiers. Output and input functions, Operators - Arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, Assignment, Conditional and type casting. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify type casting 2 applying introduction to programming concepts: structure of a c program - keyword, variables, constants, data types and type qualifiers. output and input functions, operators - arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, assignment, conditional and type casting. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Aspect	What to prepare
Definition/identity	What Identify type casting 2 Applying Introduction to Programming concepts: Structure of a C program - Keyword, Variables, Constants, Data types and type qualifiers. Output and input functions, Operators - Arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, Assignment, Conditional and type casting. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1- Identify type casting 2 Applying Introduction to Programming concepts: Structure of a C program - Keyword, Variables, Constants, Data types and type qualifiers. Output and input functions, Operators - Arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, Assignment, Conditional and type casting. definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Make use of programming concepts with C programming.

M1.01	Understand structure of a C program	1	
Understanding.			
	Use the terms Keywords, Variables, Constants,		
M1.02	Data types and type qualifiers	3	Applying
M1.03	Identify output and input functions	3	Applying
M1.04	Identify operators	4	Applying
M1.05	Identify type casting	2	Applying

Contents:

Introduction to Programming concepts: Structure of a C program - Keyword, Variables, Constants, Data types and type qualifiers. Output and input functions, Operators - Arithmetic, relational, logical, increment/decrement, conditional, assignment, bit wise, Assignment, Conditional and type casting.

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Make use of iterative control structures and arrays in programs.

This module is presented from the official syllabus scope for Programming in C. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Make use of iterative control structures and arrays in programs.
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

2 Applying nested if and switch Identify loop structures – while, do-while, for

2 Applying nested if and switch Identify loop structures – while, do-while, for is an official topic in Module 2 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying nested if and switch Identify loop structures – while, do-while, for using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying nested if and switch identify loop structures – while, do-while, for should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying nested if and switch Identify loop structures – while, do-while, for means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-2 Applying nested if and switch Identify loop structures - while, do-while, for **മലയാളം** definition/meaning **മലയാളം**. **മലയാളം** **മലയാളം** **മലയാളം**, steps, application **മലയാളം** **മലയാളം** **മലയാളം**. Technical terms English- **മലയാളം** **മലയാളം**.

2 Applying and nested loops. Make use of unconditional control statements -

2 Applying and nested loops. Make use of unconditional control statements - is an official topic in Module 2 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying and nested loops. Make use of unconditional control statements - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying and nested loops. make use of unconditional control statements - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying and nested loops. Make use of unconditional control statements - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- 2 Applying and nested loops. Make use of unconditional control statements - definition/meaning, steps, application. Technical terms English- .

2 Applying break, continue and goto Identify one dimensional array operations -

2 Applying break, continue and goto Identify one dimensional array operations - is an official topic in Module 2 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Applying break, continue and goto Identify one dimensional array operations - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 applying break, continue and goto identify one dimensional array operations - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Applying break, continue and goto Identify one dimensional array operations - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-2 Applying break, continue and goto Identify one dimensional array operations - ഏകമാത്രാവിന്യാസങ്ങൾ definition/meaning ഉപയോഗിക്കുക. ഏകമാത്രാവിന്യാസങ്ങൾ, steps, application ഉപയോഗിക്കുക. Technical terms English-ഏകമാത്രാവിന്യാസങ്ങൾ.

3 Applying. insertion, deletion, searching, sorting Use two dimensional array deceleration and

3 Applying. insertion, deletion, searching, sorting Use two dimensional array deceleration and is an official topic in Module 2 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying. insertion, deletion, searching, sorting Use two dimensional array deceleration and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying. insertion, deletion, searching, sorting use two dimensional array deceleration and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying. insertion, deletion, searching, sorting Use two dimensional array deceleration and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Applying. insertion, deletion, searching, sorting Use two dimensional array deceleration and definition/meaning . . . , steps, application Technical terms English-

3 Applying. matrix operations

3 Applying. matrix operations is an official topic in Module 2 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying. matrix operations using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying. matrix operations should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying. matrix operations means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-3 Applying. matrix operations $\square\square\square\square\square\square\square\square$
 $\square\square\square\square$ definition/meaning $\square\square\square\square\square\square\square\square\square\square$. $\square\square\square\square\square$ $\square\square\square\square\square$ $\square\square\square\square\square\square$, steps,
application $\square\square\square\square\square$ $\square\square\square\square\square\square\square\square$ $\square\square\square\square\square$. Technical terms English- $\square$ $\square\square\square\square$
 $\square\square\square\square\square\square\square\square$.

Make use of matrix manipulations 3 Applying Iterative control structures and arrays: Control structures - if, if-else, nested if statement, and switch.

Looping structures - while, do-while, for, nested loops, Unconditional control statements-break, continue, goto. Array operations - insertion, deletion, searching, sorting. Two dimensional array declaration - two dimensional array operations - Matrix manipulations (addition, multiplication etc) Develop programs using pointers and strings to solve problems more

Make use of matrix manipulations 3 Applying Iterative control structures and arrays: Control structures - if, if-else, nested if statement, and switch. Looping structures - while, do-while, for, nested loops, Unconditional control statements-break, continue, goto. Array operations - insertion, deletion, searching, sorting. Two dimensional array declaration - two dimensional array operations - Matrix manipulations (addition, multiplication etc) Develop programs using pointers and strings to solve problems more is an official topic in Module 2 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of matrix manipulations 3 Applying Iterative control structures and arrays: Control structures - if, if-else, nested if statement, and switch. Looping structures - while, do-while, for, nested loops, Unconditional control statements-break, continue, goto. Array operations - insertion, deletion, searching, sorting. Two dimensional array declaration - two dimensional array operations - Matrix manipulations (addition, multiplication etc) Develop programs using pointers and strings to solve problems more using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of matrix manipulations 3 applying iterative control structures and arrays: control structures - if, if-else, nested if statement, and switch. looping structures - while, do-while, for, nested loops, unconditional control statements-break, continue, goto. array operations - insertion, deletion, searching, sorting. two dimensional array declaration - two dimensional array operations - matrix manipulations (addition, multiplication etc) develop programs using pointers and strings to solve problems more should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of matrix manipulations 3 Applying Iterative control structures and arrays: Control structures - if, if-else, nested if statement, and switch. Looping structures - while, do-while, for, nested loops, Unconditional control statements-break, continue, goto. Array operations - insertion, deletion, searching, sorting. Two dimensional array declaration - two dimensional array operations - Matrix manipulations (addition, multiplication etc) Develop programs using pointers and strings to solve problems more means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Make use of matrix manipulations 3 Applying Iterative control structures and arrays: Control structures - if, if-else, nested if statement, and switch. Looping structures - while, do-while, for, nested loops, Unconditional control statements-break, continue, goto. Array operations - insertion, deletion, searching, sorting. Two dimensional array declaration - two dimensional array operations - Matrix manipulations (addition, multiplication etc) Develop programs using pointers and strings to solve problems more
 പരമ്പരാഗതമായി പഠിക്കുന്ന definition/meaning പരമ്പരാഗതമായി. പരമ്പരാഗതമായി പരമ്പരാഗതമായി, steps, application പരമ്പരാഗതമായി പരമ്പരാഗതമായി. Technical terms English- പരമ്പരാഗതമായി പരമ്പരാഗതമായി.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Make use of iterative control structures and arrays in programs.

Identify the control structures – if, if-else

M2.01
Applying 2

nested if and switch

Identify loop structures – while, do-while, for

M2.02
Applying 2

and nested loops.

Make use of unconditional control statements –

M2.03
Applying 2

break, continue and goto

Identify one dimensional array operations –

M2.04
Applying. 3

insertion, deletion, searching, sorting

Use two dimensional array deceleration and

M2.05
Applying. 3

matrix operations

M2.06
Applying 3

Make use of matrix manipulations

Series Test – I 1

Contents:

Iterative control structures and arrays: Control structures - if, if-else, nested

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

efficiently.

This module is presented from the official syllabus scope for Programming in C. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: efficiently.
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Use pointer concepts in programming 2 Applying Identify pointer arithmetic operations -

Use pointer concepts in programming 2 Applying Identify pointer arithmetic operations - is an official topic in Module 3 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Use pointer concepts in programming 2 Applying Identify pointer arithmetic operations - using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, use pointer concepts in programming 2 applying identify pointer arithmetic operations - should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Use pointer concepts in programming 2 Applying Identify pointer arithmetic operations - means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-പുറം Use pointer concepts in programming 2
Applying Identify pointer arithmetic operations - പൂർണ്ണസംഖ്യകളുടെ പൂർണ്ണ
definition/meaning പൂർണ്ണസംഖ്യകളുടെ. പൂർണ്ണസംഖ്യകളുടെ പൂർണ്ണസംഖ്യകളുടെ, steps, application
പൂർണ്ണസംഖ്യകളുടെ പൂർണ്ണസംഖ്യകളുടെ. Technical terms English-പൂർണ്ണസംഖ്യകളുടെ പൂർണ്ണസംഖ്യകളുടെ.

addition, subtraction, increment, decrement 3 Applying and comparison. Develop programs for accessing strings using

addition, subtraction, increment, decrement 3 Applying and comparison. Develop programs for accessing strings using is an official topic in Module 3 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify addition, subtraction, increment, decrement 3 Applying and comparison. Develop programs for accessing strings using using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, addition, subtraction, increment, decrement 3 applying and comparison. develop programs for accessing strings using should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What addition, subtraction, increment, decrement 3 Applying and comparison. Develop programs for accessing strings using means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-എന്നു ചേർക്കൽ, കുറയ്ക്കൽ, വർദ്ധന, കുറയ്ക്കൽ 3
Applying and comparison. Develop programs for accessing strings using
രേഖാവിവരങ്ങൾ രേഖാവിവരങ്ങൾ definition/meaning രേഖാവിവരങ്ങൾ. രേഖാവിവരങ്ങൾ രേഖാവിവരങ്ങൾ, steps, application രേഖാവിവരങ്ങൾ രേഖാവിവരങ്ങൾ രേഖാവിവരങ്ങൾ. Technical terms English-എന്നു ചേർക്കൽ
രേഖാവിവരങ്ങൾ.

3 Applying pointers

3 Applying pointers is an official topic in Module 3 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying pointers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying pointers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying pointers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-3 Applying pointers ഫുണ്ടമെന്റൽ ഫുണ്ടമെന്റൽ definition/meaning ഫുണ്ടമെന്റൽ. ഫുണ്ടമെന്റൽ ഫുണ്ടമെന്റൽ, steps, application ഫുണ്ടമെന്റൽ ഫുണ്ടമെന്റൽ ഫുണ്ടമെന്റൽ. Technical terms English-3 ഫുണ്ടമെന്റൽ ഫുണ്ടമെന്റൽ.

Utilize gets() and puts() functions 3 Applying Use built in string handling functions like

Utilize gets() and puts() functions 3 Applying Use built in string handling functions like is an official topic in Module 3 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Utilize gets() and puts() functions 3 Applying Use built in string handling functions like using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, utilize gets() and puts() functions 3 applying use built in string handling functions like should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Utilize gets() and puts() functions 3 Applying Use built in string handling functions like means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഉൾ ഉപയോഗിക്കുക gets() and puts() functions 3 Applying Use built in string handling functions like `strlen()` `strcpy()` definition/meaning ഉപയോഗിക്കുക. `strlen()` `strcpy()` `strcat()`, steps, application ഉപയോഗിക്കുക ഉപയോഗിക്കുക. Technical terms English-ൽ ഉപയോഗിക്കുക ഉപയോഗിക്കുക.

4 Applying strcpy(), strlen(), strcat(), strcmp(), strrev() Pointers and string operations: Basics of Pointers, Pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and Puts() functions, String handling functions.

4 Applying strcpy(), strlen(), strcat(), strcmp(), strrev() Pointers and string operations: Basics of Pointers, Pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and Puts() functions, String handling functions. is an official topic in Module 3 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Applying strcpy(), strlen(), strcat(), strcmp(), strrev() Pointers and string operations: Basics of Pointers, Pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and Puts() functions, String handling functions. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 applying strcpy(), strlen(), strcat(), strcmp(), strrev() pointers and string operations: basics of pointers, pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and puts() functions, string handling functions. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
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Aspect	What to prepare
Definition/identity	What 4 Applying strcpy(), strlen(), strcat(), strcmp(), strcmp() Pointers and string operations: Basics of Pointers, Pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and Puts() functions, String handling functions. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-4 Applying strcpy(), strlen(), strcat(), strcmp(), strcmp() Pointers and string operations: Basics of Pointers, Pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and Puts() functions, String handling functions. definition/meaning. steps, application. Technical terms English- Malayalam.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

efficiently.

M3.01	Use pointer concepts in programming	2
Applying		
M3.02	Identify pointer arithmetic operations - addition, subtraction, increment, decrement	3
Applying		
M3.03	and comparison. Develop programs for accessing strings using	3
Applying		
M3.04	pointers Utilize gets() and puts() functions	3
Applying		
M3.05	Use built in string handling functions like	4
Applying		
	strcpy(), strlen(), strcat(), strcmp(), strrev()	
Contents:		
Pointers and string operations: Basics of Pointers, Pointers arithmetic operations - addition, subtraction, increment, decrement, comparison, accessing strings using pointers, gets() and Puts() functions, String handling functions.		

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

Make use of functions to solve problems effectively

This module is presented from the official syllabus scope for Programming in C. Use the topic cards for learning and the source transcript for exact coverage.

As specified

CO-linked

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Make use of functions to solve problems effectively
- Official duration: As specified
- Official topic entries captured: 6
- The full source excerpt is preserved below for coverage checking.

Explain functions, types and advantages

Explain functions, types and advantages is an official topic in Module 4 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain functions, types and advantages using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain functions, types and advantages should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain functions, types and advantages means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓം Explain functions, types and advantages
ഓംഓംഓംഓംഓംഓം ഓംഓം definition/meaning ഓംഓംഓംഓംഓംഓം. ഓംഓംഓം ഓംഓംഓം ഓംഓംഓം,
steps, application ഓംഓംഓം ഓംഓംഓംഓംഓം ഓംഓംഓം. Technical terms English-ഓം ഓംഓംഓം
ഓംഓംഓംഓംഓം.

Illustrate structure of a user defined functions 3 Understanding Make use of function prototype, function call,

Illustrate structure of a user defined functions 3 Understanding Make use of function prototype, function call, is an official topic in Module 4 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate structure of a user defined functions 3 Understanding Make use of function prototype, function call, using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate structure of a user defined functions 3 understanding make use of function prototype, function call, should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate structure of a user defined functions 3 Understanding Make use of function prototype, function call, means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓരോ ഉദാഹരണത്തിനും ഉപയോഗിക്കാവുന്ന 3 Understanding Make use of function prototype, function call, ഫങ്ഷൻ പ്രോട്ടോടൈപ്പ് ഫങ്ഷൻ definition/meaning ഫങ്ഷൻ പ്രോട്ടോടൈപ്പ്. ഫങ്ഷൻ പ്രോട്ടോടൈപ്പ്, steps, application ഫങ്ഷൻ പ്രോട്ടോടൈപ്പ് ഫങ്ഷൻ. Technical terms English-ഓരോ ഉദാഹരണത്തിനും ഉപയോഗിക്കാവുന്ന 3

3 Applying arguments, return type

3 Applying arguments, return type is an official topic in Module 4 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying arguments, return type using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying arguments, return type should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying arguments, return type means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Applying arguments, return type
പ്രതികരണങ്ങൾക്ക് definition/meaning നൽകുന്നു. പ്രതികരണങ്ങൾക്ക്, steps, application നൽകുന്നു. Technical terms English-ൽ പ്രതികരണങ്ങൾക്ക്.

Identify different types of functions 2 Applying Make use of call by value and call by

Identify different types of functions 2 Applying Make use of call by value and call by is an official topic in Module 4 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Identify different types of functions 2 Applying Make use of call by value and call by using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, identify different types of functions 2 applying make use of call by value and call by should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Identify different types of functions 2 Applying Make use of call by value and call by means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓക്സ്ജനം Identify different types of functions 2 Applying Make use of call by value and call by ഓക്സ്ജനം definition/meaning ഓക്സ്ജനം. ഓക്സ്ജനം ഓക്സ്ജനം, steps, application ഓക്സ്ജനം ഓക്സ്ജനം. Technical terms English-ഓക്സ്ജനം ഓക്സ്ജനം.

3 Applying reference

3 Applying reference is an official topic in Module 4 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Applying reference using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 applying reference should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Applying reference means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-3 Applying reference ഓരോന്നിനും നിർവ്വചനം/അർത്ഥം ഉണ്ടാകണം. ഓരോന്നിനും ഓരോന്നിനും, steps, application ഉണ്ടാകണം. Technical terms English-ൽ ഉണ്ടാകണം.

Make use of recursive functions 2 Applying Functions: Definition of Functions
- Structure of user defined functions - functionprototype, function call, arguments, return type - Passing arguments to a Function: call byvalue, call by reference, Recursive functions. Text / Reference: T/R Book Title/Author T1 Programming in ANSI C by Balaguruswami E, TMH publications Computing Fundamentals and C Programming | 2nd Edition by E R1 Balaguruswami, TMH publishers R2 Programming in C - by ReemaThareja, OUP India publishers C Programming Absolute Beginner's Guide by Greg Perry, Dean Miller, QUE R3 Publications R4 C: The Complete Reference by Herbert Schildt, TMH Publications R5 Programming in C - second edition - R. Subburaj, - Vikas Publishing House. Online Resources: Sl.No Website Link 1 <https://www.programiz.com/c-programming> 2 <https://www.tutorialspoint.com/cprogramming/index.htm> 3 <https://www.cprogramming.com/> 4 <https://www.geeksforgeeks.org/c-programming-language/> 5 <https://www.freecodecamp.org/news/the-c-beginners-handbook/>

Make use of recursive functions 2 Applying Functions: Definition of Functions - Structure of user defined functions - functionprototype, function call, arguments, return type - Passing arguments to a Function: call byvalue, call by reference, Recursive functions. Text / Reference: T/R Book Title/Author T1 Programming in ANSI C by Balaguruswami E, TMH publications Computing Fundamentals and C Programming | 2nd Edition by E R1 Balaguruswami, TMH publishers R2 Programming in C - by ReemaThareja, OUP India publishers C Programming Absolute Beginner's Guide by Greg Perry, Dean Miller, QUE R3 Publications R4 C: The Complete Reference by Herbert Schildt, TMH Publications R5 Programming in C - second edition - R. Subburaj, - Vikas Publishing House. Online Resources: Sl.No Website Link 1 <https://www.programiz.com/c-programming> 2 <https://www.tutorialspoint.com/cprogramming/index.htm> 3 <https://www.cprogramming.com/> 4 <https://www.geeksforgeeks.org/c-programming-language/> 5 <https://www.freecodecamp.org/news/the-c-beginners-handbook/> is an official topic in Module 4 of Programming in C. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Make use of recursive functions 2 Applying Functions: Definition of Functions - Structure of user defined functions - functionprototype, function call, arguments, return type - Passing arguments to a Function: call byvalue, call by reference, Recursive functions. Text / Reference: T/R Book Title/Author T1 Programming in ANSI C by Balaguruswami E, TMH publications Computing Fundamentals and C Programming | 2nd Edition by E R1 Balaguruswami, TMH publishers R2 Programming in C - by ReemaThareja, OUP India publishers C Programming Absolute Beginner's Guide by Greg Perry, Dean Miller, QUE R3 Publications R4 C: The Complete Reference by Herbert Schildt, TMH Publications R5 Programming in C - second edition - R. Subburaj, - Vikas Publishing House. Online Resources: Sl.No Website Link 1 <https://www.programiz.com/c-programming> 2 <https://www.tutorialspoint.com/cprogramming/index.htm> 3 <https://www.cprogramming.com/> 4 <https://www.geeksforgeeks.org/c-programming-language/> 5 <https://www.freecodecamp.org/news/the-c-beginners-handbook/> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, make use of recursive functions 2 applying functions: definition of functions - structure of user defined functions - functionprototype, function call, arguments, return type - passing arguments to a function: call byvalue, call by reference, recursive functions. text / reference: t/r book title/author t1 programming in ansi c by balaguruswami e, tmh publications computing fundamentals and c programming | 2nd edition by e r1 balaguruswami, tmh publishers r2 programming in c - by reemathareja, oup india publishers c programming absolute beginner's guide by greg perry, dean miller, que r3 publications r4 c: the complete reference by herbert schildt, tmh publications r5 programming in c - second edition - r. subburaj, - vikas publishing house. online resources: sl.no website link 1 <https://www.programiz.com/c-programming> 2 <https://www.tutorialspoint.com/cprogramming/index.htm> 3 <https://www.cprogramming.com/> 4 <https://www.geeksforgeeks.org/c-programming-language/> 5 <https://www.freecodecamp.org/news/the-c-beginners-handbook/> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Make use of recursive functions 2 Applying Functions: Definition of Functions - Structure of user defined functions - functionprototype, function call, arguments, return type - Passing

Aspect	What to prepare
	arguments to a Function: call byvalue, call by reference, Recursive functions. Text / Reference: T/R Book Title/Author T1 Programming in ANSI C by Balaguruswami E, TMH publications Computing Fundamentals and C Programming 2nd Edition by E R1 Balaguruswami, TMH publishers R2 Programming in C - by ReemaThareja, OUP India publishers C Programming Absolute Beginner's Guide by Greg Perry, Dean Miller, QUE R3 Publications R4 C: The Complete Reference by Herbert Schildt, TMH Publications R5 Programming in C - second edition - R. Subburaj, - Vikas Publishing House. Online Resources: Sl.No Website Link 1 https://www.programiz.com/c-programming 2 https://www.tutorialspoint.com/cprogramming/index.htm 3 https://www.cprogramming.com/ 4 https://www.geeksforgeeks.org/c-programming-language/ 5 https://www.freecodecamp.org/news/the-c-beginners-handbook/ means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4- Make use of recursive functions 2 Applying Functions: Definition of Functions - Structure of user defined functions - function prototype, function call, arguments, return type - Passing arguments to a Function: call byvalue, call by reference, Recursive functions. Text / Reference: T/R Book Title/Author T1 Programming in ANSI C by Balaguruswami E, TMH publications Computing Fundamentals and C Programming | 2nd Edition by E R1 Balaguruswami, TMH publishers R2 Programming in C - by ReemaThareja, OUP India publishers C Programming Absolute Beginner's Guide by Greg Perry, Dean Miller, QUE R3 Publications R4 C: The Complete Reference by Herbert Schildt, TMH Publications R5 Programming in C - second edition - R. Subburaj, - Vikas Publishing House. Online Resources: Sl.No Website Link 1 <https://www.programiz.com/c-programming> 2 <https://www.tutorialspoint.com/cprogramming/index.htm> 3 <https://www.cprogramming.com/> 4 <https://www.geeksforgeeks.org/c-programming-language/> 5 <https://www.freecodecamp.org/news/the-c-beginners-handbook/>
 ഫങ്ഷന്റെ ഓരോ ഭാഗവും വിശദീകരിക്കുക. ഫങ്ഷന്റെ ഓരോ ഭാഗവും വിശദീകരിക്കുക, steps, application
 ഫങ്ഷന്റെ ഓരോ ഭാഗവും വിശദീകരിക്കുക. Technical terms English-
 ഫങ്ഷന്റെ ഓരോ ഭാഗവും വിശദീകരിക്കുക.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Make use of functions to solve problems effectively

M4.01	Explain functions, types and advantages	2
-------	---	---

Understanding

M4.02	Illustrate structure of a user defined functions	3
-------	--	---

Understanding

M4.03	Make use of function prototype, function call, arguments, return type	3
-------	---	---

Applying

M4.04	Identify different types of functions	2
-------	---------------------------------------	---

Applying

M4.05	Make use of call by value and call by reference	3
-------	---	---

Applying

M4.06	Make use of recursive functions	2
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Applying

	Series Test – II	1
--	------------------	---

Contents:

Functions: Definition of Functions - Structure of user defined functions - function prototype, function call, arguments, return type - Passing arguments to a Function:

call by value, call by reference, Recursive functions.

Text / Reference:

T/R

Book Title/Author

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Understand structure of a C program 1 Understanding. Use the terms Keywords, Variables, Constants,. (3 marks)

Answer: Begin with the standard definition or identity of *Understand structure of a C program 1 Understanding*. Use the terms *Keywords, Variables, Constants,*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying Data types and type qualifiers. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying Data types and type qualifiers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify output and input functions. (3 marks)

Answer: Begin with the standard definition or identity of *Identify output and input functions*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify operators. (3 marks)

Answer: Begin with the standard definition or identity of *Identify operators*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Applying nested if and switch Identify loop structures - while, do-while, for. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Applying nested if and switch Identify loop structures - while, do-while, for*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Applying and nested loops. Make use of unconditional control statements -. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Applying and nested loops. Make use of unconditional control statements -*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 2 Applying break, continue and goto Identify one dimensional array operations -. (3 marks)

Answer: Begin with the standard definition or identity of 2 *Applying break, continue and goto Identify one dimensional array operations -*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying. insertion, deletion, searching, sorting Use two dimensional array deceleration and. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying. insertion, deletion, searching, sorting Use two dimensional array deceleration and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Use pointer concepts in programming 2 Applying Identify pointer arithmetic operations -. (3 marks)

Answer: Begin with the standard definition or identity of *Use pointer concepts in programming 2 Applying Identify pointer arithmetic operations -.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain addition, subtraction, increment, decrement 3 Applying and comparison. Develop programs for accessing strings using. (3 marks)

Answer: Begin with the standard definition or identity of *addition, subtraction, increment, decrement 3 Applying and comparison. Develop programs for accessing strings using*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying pointers. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying pointers*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Utilize gets() and puts() functions 3 Applying Use built in string handling functions like. (3 marks)

Answer: Begin with the standard definition or identity of *Utilize gets() and puts() functions 3 Applying Use built in string handling functions like*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Explain functions, types and advantages. (3 marks)

Answer: Begin with the standard definition or identity of *Explain functions, types and advantages*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Illustrate structure of a user defined functions 3 Understanding Make use of function prototype, function call,. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate structure of a user defined functions 3 Understanding Make use of function prototype, function call,.* State

the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 3 Applying arguments, return type. (3 marks)

Answer: Begin with the standard definition or identity of *3 Applying arguments, return type*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Identify different types of functions 2 Applying Make use of call by value and call by. (3 marks)

Answer: Begin with the standard definition or identity of *Identify different types of functions 2 Applying Make use of call by value and call by*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and

marks.

Module 1

1-mark definition: State the meaning of **Understand structure of a C program 1 Understanding. Use the terms Keywords, Variables, Constants,.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Applying nested if and switch Identify loop structures - while, do-while, for.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Use pointer concepts in programming 2 Applying Identify pointer arithmetic operations -.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Explain functions, types and advantages.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link <https://www.programiz.com/c-programming>
- <https://www.tutorialspoint.com/cprogramming/index.htm>
<https://www.cprogramming.com/>
- <https://www.geeksforgeeks.org/c-programming-language/>
- <https://www.freecodecamp.org/news/the-c-beginners-handbook/>

IN-PAGE SEARCH

Search this lesson

Prepared from the supplied official SITTTT syllabus PDF for Course 5243B. Verify institutional updates against the current official curriculum.